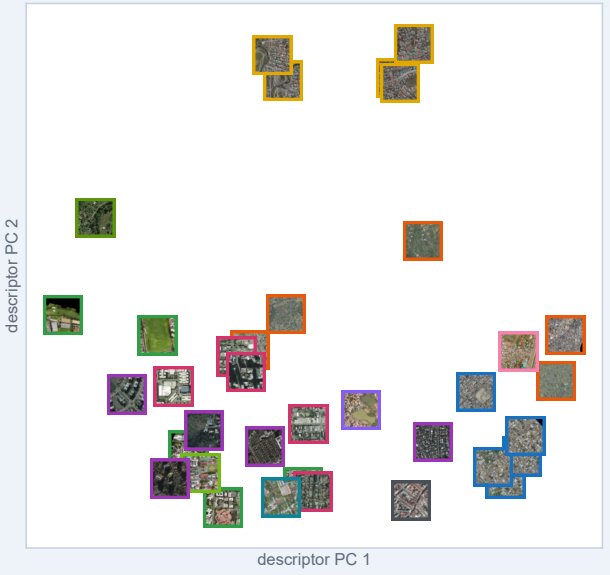
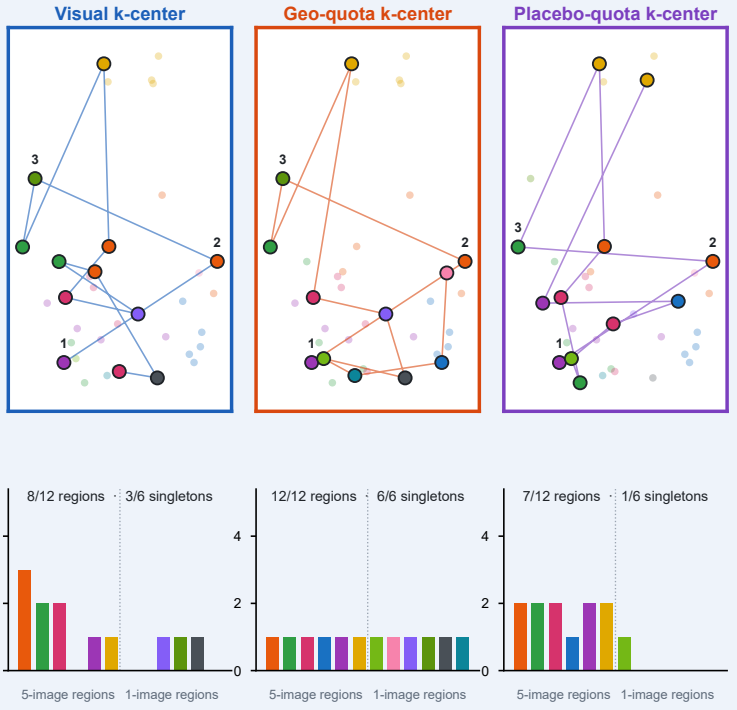
a Skewed 36-image candidate pool in DINOv2 space



Accra · Bogotá · Chicago · Pointe-Noire · Santiago · Ulaanbaatar
Mahé · Maputo · Niamey · Rio · Vienna · Dar es Salaam

Squares = candidates per source region: six regions offer five images, six offer only one

b Same pool, three rules, 12 picks each (recorded runs)



list fixed → masks released

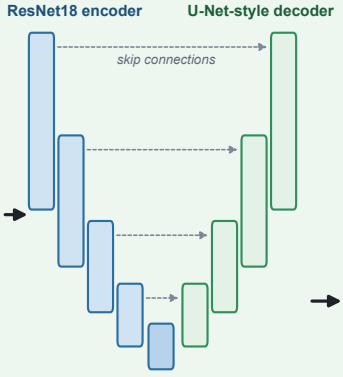
mask firewall

c Only the chosen list crosses the firewall



Geo-quota k-center list: one image from each of the 12 source regions

Bareland · Rangeland · Developed · Road
Tree · Water · Agriculture · Building



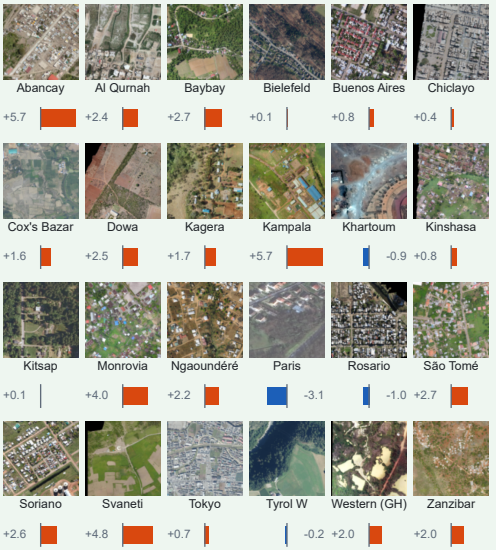
all 12.46 M parameters trained
224-px crops, flips, quarter rotations

control: frozen DINOv2 patch features + 128-unit MLP head (budgets 6, 12, 24)

Budget: 12 whole images per fit. Calibration (24 images) and source validation (20 images) use shared development labels outside this budget.

d Scored in 24 regions never used for fitting

Geo-quota k-center − visual k-center: +1.68 mIoU points
95% region bootstrap [+0.86, +2.51] · higher in 20 of 24 regions



bars under tiles: regional difference · orange = quota better, blue = k-center better